

Automated Item Generation with Generative and Predictive ML

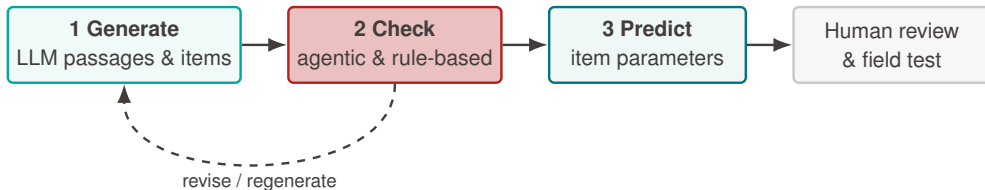
Fall School, University of Tübingen — Day 2: Screening, prediction, artificial test-takers

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Session 3A — Stage 2: Rule-based and agent-based screening



Morning: screening (rules, SAQUET, sensibility/solvability) → predicting parameters from item text. Notebook 3 (Parts A and B).

Afternoon: LLMs as artificial test-takers → validating the framework → what transfers. Notebook 4.

Input for everything today: your `day1_items.json`.

One item from yesterday's files (on the blackboard): would you put it in a high-stakes test?

List the objections — they become this morning's screening criteria.

Cheap checks first

Which of the following is **not** a reason the author gives for the decline of the harbour?

- A. silting
- B. war
- C. the railway, which redirected trade to the inland cities
- D. all of the above

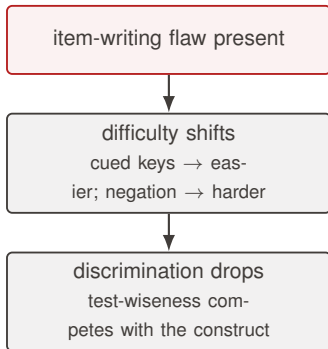
negation in stem

longest option is the key

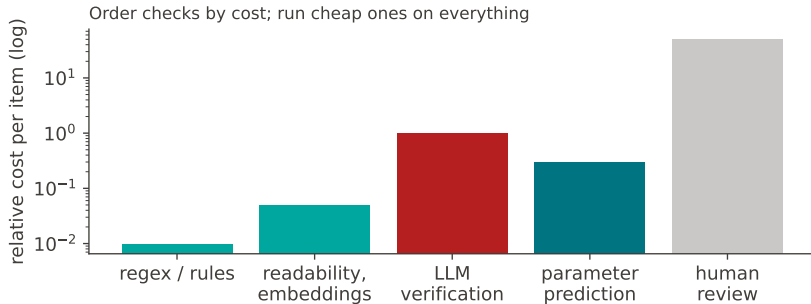
"all of the above"

Haladyna's guidelines; Tarrant et al. (2006): a 19-criteria rubric that most later work, including SAQUET, adopts.

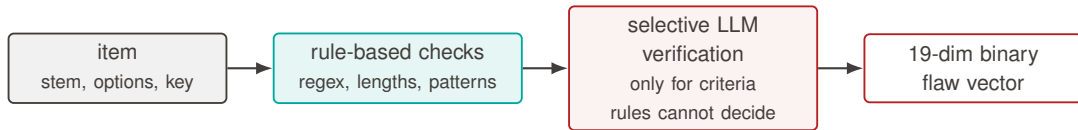
Flaws are construct-irrelevant: test-wise examinees exploit them, others are penalised.



Recent work links IWF presence directly to IRT parameter estimates: flawed items behave differently in calibration, not just in expert judgement. This is the bridge to Block B: if flaws move parameters, screening them out before prediction removes noise from the predictor.



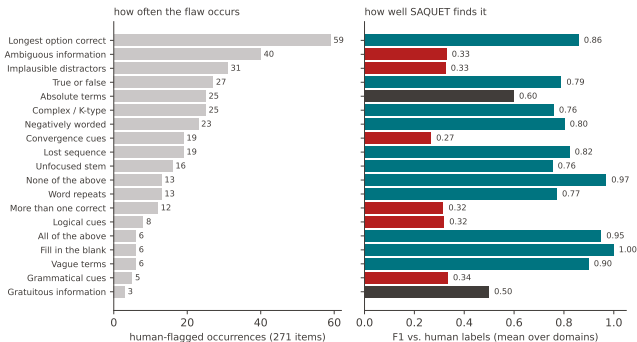
Regex and rules on every item; an LLM only where a rule cannot decide; parameter prediction once; humans last. The framework's order is an economic order.



Open-source toolkit implementing the Tarrant rubric; hybrid design; >94% agreement with human raters on whether an item has any flaw.

Moore et al. 2024

What SAQUET catches well and badly



The formal flaws (none/all of the above, fill in the blank) are found almost perfectly — but they are rare, so the check rarely changes anything.

The frequent flaws are where a screen earns its keep: *longest option correct* is detected well; *ambiguous information* and *implausible distractors* are not.

SAQUET is a screen, not a judge.

Moore et al. (2024), Table 3; F1 averaged over the domains in which humans flagged the flaw.

sensibility

is the question well-
formed on its own?

solvability, factual mode

must be answerable
only with the passage

solvability, literary mode

must be answerable
from the passage

implemented as two model calls per item: without the passage, then with it; the pattern of correct answers decides

Jensen (2025) §7.1; Jensen (2026) §3.1

length parity
key vs. distractors

overlap with key
string and embedding

semantic distance
among options

attractiveness proxies
learned scor-
ers (DISTO)

These proxies are the best we have until this afternoon, when responses exist and distractor functioning can be measured directly.

Jensen (2025) §6.3

can: sensitive-content lexicons, LLM sensitivity review against published guidelines (e.g., DET fair-content guidelines)

cannot: differential item functioning — needs response data by group

This is not a DIF workshop; the pointer is that DIF analysis belongs in the validation block, after response data exist (this afternoon).



compare rule-only vs. rule + LLM flags on the same items; stretch: write one new rule (key-length z-score) and see what it flags

Notebook by R. Jensen

- Which flags fired most often?
- Where did the LLM verifier disagree with the rules?

We now have items that *look* fine. Are they *hard*?

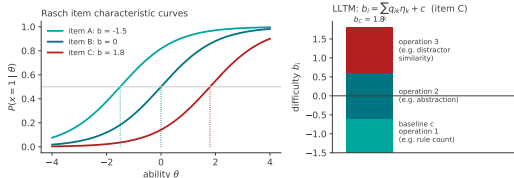
Coffee — back at 10:45

Session 3B — Stage 3a:

Predicting item parameters from text



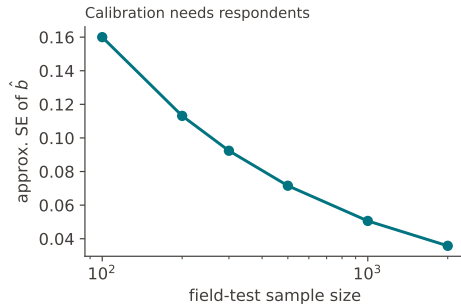
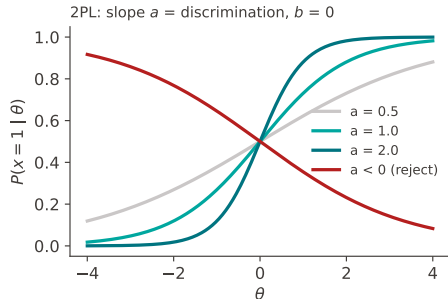
Difficulty before data



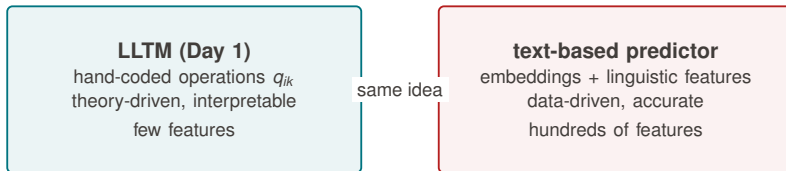
$$P(x_{pi} = 1 \mid \theta_p, b_i) = \frac{e^{\theta_p - b_i}}{1 + e^{\theta_p - b_i}}$$

b_i : the ability at which the item is solved with probability .5.

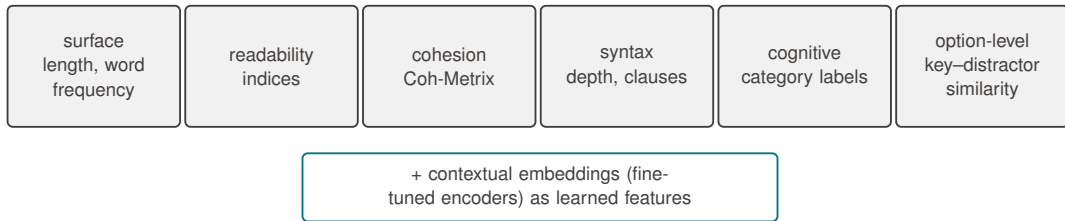
- person and item on one scale
- difficulty is a *population-referenced* quantity



$\hat{a} > 0$ is a hard filter in the framework; MCQs may also need a guessing parameter (3PL). Standard errors shrink with $\sqrt{n}$ — hundreds of respondents per item is the bottleneck the predictor tries to shortcut.



Trade-off: interpretability vs. accuracy. Both assume that what makes an item hard is visible in the item.



Systematic reviews: AlKhuzaei et al. (2023); Li et al. Typical ceilings: $r \approx .5$ –.8 for difficulty, much lower for discrimination.

667 retired USMLE items; 1PL difficulty and response time

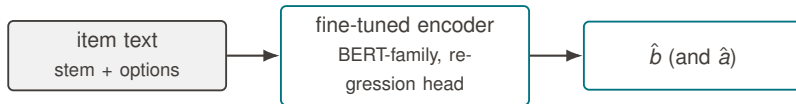
difficulty harder to predict than response time

best: ensembles of embeddings + linguistic features

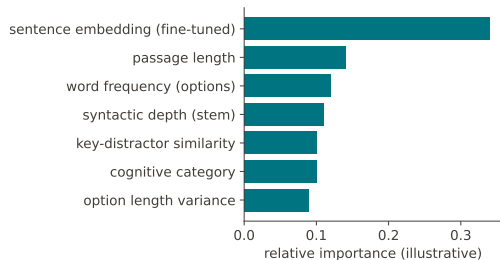
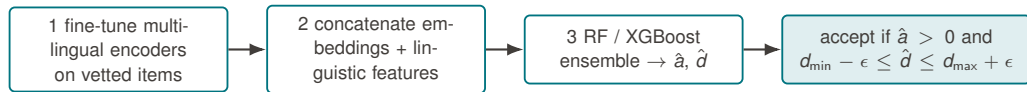
LLM-generated answers as extra features helped

A rare benchmark with real item parameters — and a sobering one: RMSE spread across systems was small, and the gap to the baseline modest.

Yaneva et al. 2024



Gombert et al. (2024): scalar-mixed encoders; Li et al. (2025): fine-tuned small and large LMs; state of the art still means $r \approx .5-.8$



Validation on the human item bank: $r \approx .78$ predicted vs. empirical difficulty (Jensen 2025, §7.2). Feature importances here are illustrative of the typical pattern: the embedding dominates, then length and lexical features. pipeline: Jensen (2026) §3.3

passage: length, lexical rarity, cohesion

question: process type (locate / infer /
integrate / evaluate)

stem–passage distance: paraphrase
vs. verbatim

options: plausibility, overlap with pas-
sage

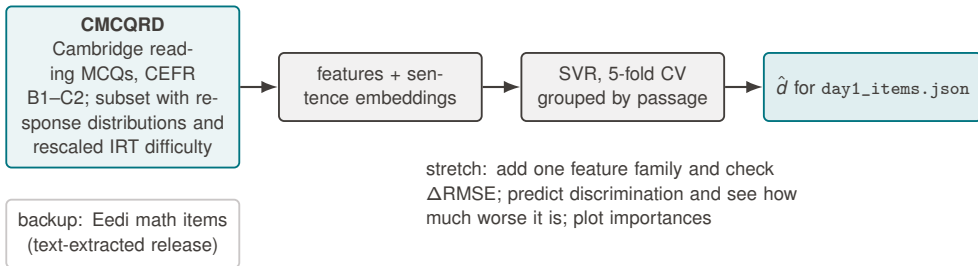
Kapoor et al. (2025, Domingue group):
an annotated repository of reading items
with human-coded features — which
ones carry signal for comprehension
difficulty.

discrimination is much harder than difficulty from text alone; it depends on the rest of the test and on metadata

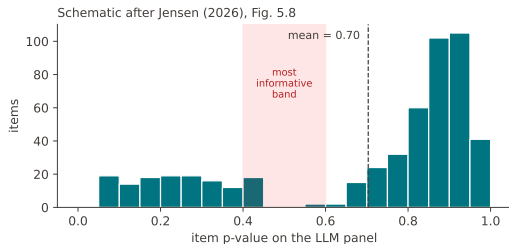
parameters are population-referenced: the predictor learns one population's scale (this afternoon's point)

domain shift: training bank vs. generated items; distribution of generated items is narrower

Han et al. 2025



Notebook by R. Jensen



Where do your generated items sit on the predicted scale?

Expect them to cluster at the easy end — the pattern Jensen (2026, Fig. 5.8) found in the empirical p-values of generated items on an LLM panel (pool means .73 and .74, more than half of the items above .8). The plot is a schematic drawn to those statistics, not the original figure. Keep it in mind for this afternoon.

Lunch — back at 13:00

Session 4A — Stage 3b: LLMs as artificial test-takers

Field-testing without a field

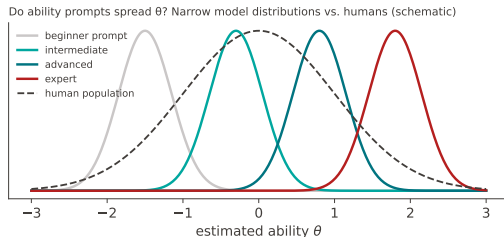


the objection: whose ability is being measured? Difficulty is population-referenced — a model panel is a population, but not the target one.

Lalor, Wu & Yu (2019): “artificial crowds” — many trained models as respondents to fit IRT on NLP test sets

person-fit and model-fit ideas transferred from psychometrics to model evaluation

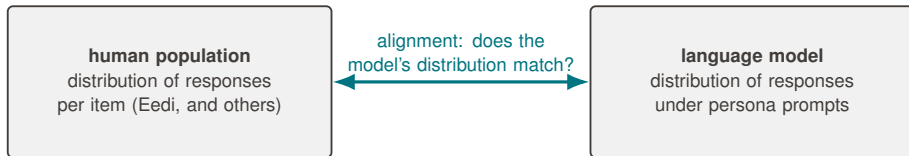
The idea predates LLMs: use a population of systems where a population of people is unavailable, and read the item parameters off the result.



Six LLMs on college algebra items:

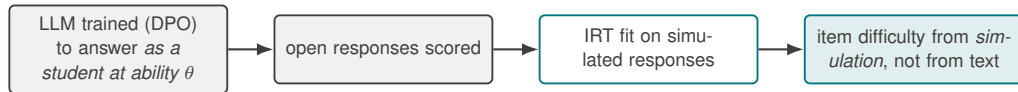
- some models at or above student proficiency
- but *narrow* proficiency distributions
- IRT difficulties from GPT-3.5/4 responses correlated $> .7$ with human-based estimates

BJET



Persona prompting helps only partly. The text-extracted Eedi release from this work is our backup dataset.

Domingue / Goodman group

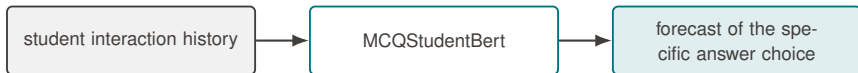


A different route to stage 3: instead of predicting parameters from item text, simulate students and calibrate. The two routes can be compared on the same items — Notebook 4.

“Take out your calculators” (2026):
prompt LLMs at several skill levels, fit
IRT, estimate MCQ math difficulty

Maeda (2025): field-testing grammar
items with AI examinees

Common design: ability is induced by *prompt*. In our reference run that alone left every persona at 20/20 — so Notebook 4 also varies the *model size* (0.6B–30B), like the model panel of Liu, Bhandari & Pardos (2025). The open question is whether induced ability spreads θ the way a classroom does.



Gado et al. (2024, Käser group): forecasting the chosen option rather than correctness. Distractor-level prediction is an alternative to simulating whole test-takers — and a direct check on whether distractors attract the students they were written for.

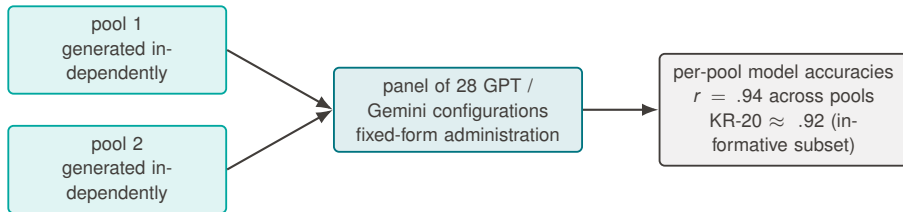
Do LLM responses look psychometrically plausible?

item-level agreement: partial — difficulty rank orders overlap with humans

discrimination structure: models fail to reproduce it

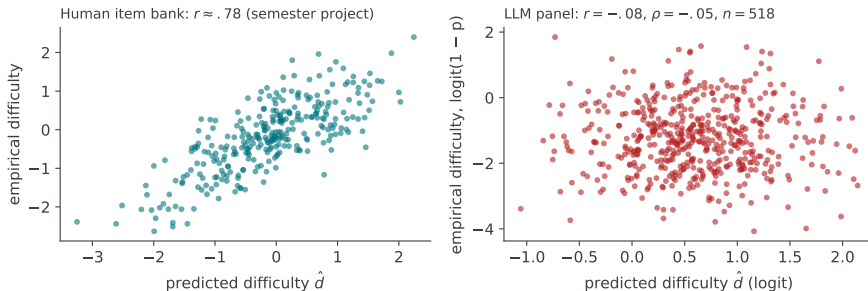
data: NAEP and CMC-QRD — the same reading set we used this morning

Domingue group, 2025



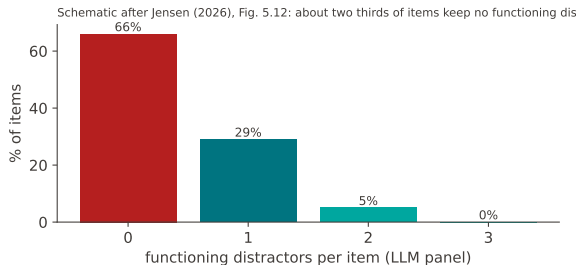
Two independently generated pools behave as parallel forms for the model panel — the generator is consistent.

Jensen (2026) ch. 4–5



Schematic reproductions: left after the validation reported in Jensen (2025), right after Jensen (2026), Fig. 5.13 — not the original data points

The human-calibrated predictor does not order items for LLM respondents: $r = -.08$ pooled, and the two pools even disagree on the sign ($-.21$ vs. $+.09$). Reasoning budget barely mattered. Jensen (2026), §5.7, Figs. 5.13–5.15



For the model panel, about two thirds of the items kept no functioning distractor and only 4–5% kept two: models either solve the item or fail on the key, but rarely fall for the intended misreading. Uniform across the four categories (Jensen 2026, §5.6, Fig. 5.12, App. A.2).

Distractors written for human misreadings are not distractors for LLMs.

two populations, two scales

difficulty is population-referenced; the predictor learned the human scale

generator not tuned to LLMs

items and distractors target human misreadings

narrow panel

28 configurations, small θ spread; categories correlate only $\rho \approx .4-.75$

what would separate these: a panel with wider induced ability; distractors mined from LLM errors; a predictor calibrated on LLM data

Jensen (2026) ch. 6–7

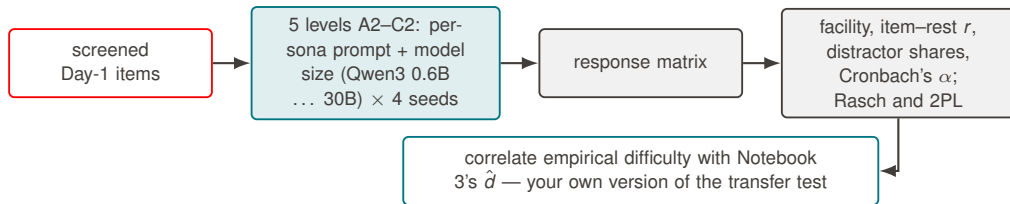
separate the *generator* model from the *respondent* panel

vary ability by prompt or model size — and check that θ actually spreads

report distractor functioning, not only accuracy

never calibrate a human test on LLM data without a transfer check

drawn from Jensen (2026) ch. 6–8



Rasch and 2PL fitted with an EM algorithm written out in numpy. Three dials: persona prompt, model size, seed.
Look at the matrix before fitting anything.

Notebook by R. Jensen

does induced
ability spread?

do distractors
get chosen?

does $\hat{d}$ rank items
the same way?

which category be-
haves differently?

Coffee — back at 14:30

Session 4B — Notebook 4, validating the framework, outlook

are the ten math isomorphs statistically isomorphic? facility per item, misconception shares

drop the C2 test-takers (the model that wrote the items) and refit: does the item order change?

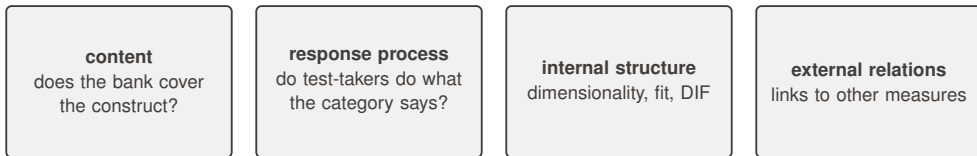
did the screening predict behaviour? passed vs. flagged items: facility, item-rest r

All three run offline from the cache (end of Notebook 4: *four more looks at the matrix*). Live demo on the endpoint: a persona with one specific misconception — does it pick the intended distractor?

Everyone replays the same run, so the numbers are shared. Each pair takes one and explains it.

- **9 of 10** reading items are solved by every test-taker, even Qwen3-0.6B — so $r(\hat{d}, b)$ is undefined for reading. What does that say about LLM respondents as stand-ins for learners?
- The ten math isomorphs range in facility from **.20 to .80**. Are they isomorphic?
- Functioning distractors: **1 of 10** reading items vs. **10 of 10** math items.
- **9 of 20** keys sit on A, only **1** on C — and the 0.6B model answers A **64%** of the time.
- The screening flagged **every** math item for grammatical cues. Real cues, or a rubric written for verbal items?

What counts as evidence that AIG worked?



Validity evidence for generated items follows the same argument as for human items. AIG changes *who* wrote the item, not what must be shown.

Jensen (2025): expert review of linguistic quality, content validity, category fit

Duolingo: large pilots with external reviewers for quality, fairness and bias, audio

Treat reviewer ratings as measurements: report agreement, calibrate automated raters against them, and keep the disagreements — they show where the automated stages are blind.

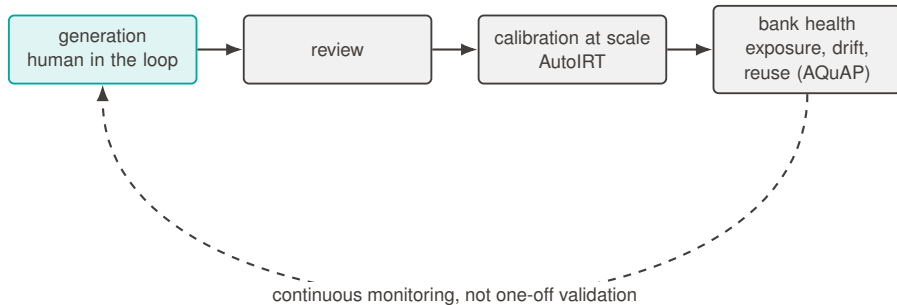
Jensen (2025) §8; Duolingo whitepapers



calibrated anchor items (A) embedded among generated items (G) → common scale

- sample sizes as for any calibration; parallel-form checks between independently generated pools (Jensen 2026: $r = .94$)
- statistical isomorphism revisited from Day 1: do template siblings share parameters?

Once response data by group exist, DIF analysis is the first thing to run on generated items — LLM-written content can carry systematic group effects that no text screen catches. Standard DIF methods apply unchanged; this school does not cover them.



failure	seen in	responsible stage
distractor collapse	Notebook 4, Jensen (2026)	1 (distractor design), 2 (proxies)
difficulty ceiling	Notebook 3, Jensen (2026)	3 (band filter), 1 (targeted generation)
category heterogeneity	Jensen (2026)	1 (constraints), validation
non-transfer of predicted difficulty	Notebook 4, Jensen (2026)	3 (population-specific calibration)
factual errors	Day 1	1 (fact check)
contamination	benchmarks	1 (open training data)

predictors calibrated
on LLM responses

adversarial distractor
mining to move items
into the .4–.6 band

difficulty-targeted gener-
ation (Uto et al. 2023)

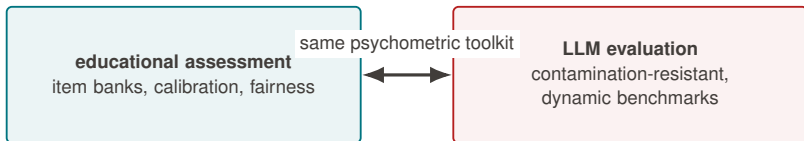
multidimensional IRT
across categories;
multimodal items

Jensen (2026) §8.1

LLM-generated personality and
situational-judgement items

network-psychometric selection of gen-
erated items (AI-GENIE)

What carries over: generation,
screening, redundancy control. What
does not: difficulty prediction — there is
no key, and the “parameter” is a loading,
not a location.



AI built with open, documented models is one way to make benchmarks that are both fresh and interpretable.

Jensen (2026) ch. 1–2

Framework and history

Gierl & Lai (2013), ITEMS module · von Davier (2018)
· Attali et al. (2022) · Säuberli & Clematide (2024) ·
Jensen (2025, 2026) — see next slide

Screening

Moore et al. (2024), SAQUET · Tarrant et al. (2006)

Prediction and simulation

Yaneva et al. (2024), BEA shared task · He-Yueya et
al. (2024) · Liu, Bhandari & Pardos (2025) · Scarlatos
et al. (2025), SMART · Gado et al. (2024),
MCQStudentBert

Data and tools

CMCQRD · Eedi (text-extracted release) · py-irt ·
Apertus · Qwen3.x

- Jensen, R. (2025). *A holistic approach to automatic item generation*. Semester project report, ETH Zurich. Supervised by R. Debelak. — the pipeline, prompts, model specialisation, sensibility/solvability, SAQUET integration, the 2PL predictor ($r \approx .78$), expert review.
- Jensen, R. (2026). *Automated Item Generation for LLM Benchmarking*. Master's thesis, Department of Computer Science, ETH Zurich. Advisors: M. Sachan, R. Debelak. — CEFR-level generation, the cognitive categories and constraints, the refinement loop, the LLM-panel benchmark, distractor collapse, non-transfer of predicted difficulty.
- The four notebooks were written by R. Jensen for this school and adapted for the endpoint and datasets used here.

the three-stage
framework: gener-
ate, check, predict

generation is cheap;
quality assur-
ance is the work

text predicts difficulty
for humans partly,
for models poorly

every automated
check needs a human
calibration point

exit poll
pace, level, one change

notebooks and slides
course page

contact
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Thank you